

**VALIDATION REPORT:
PHOTON ATTENUATION SIMULATOR
(V1.09)**

Prepared by: Cormac Thomas

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Affiliation: University of New Mexico B.S. Nuclear Engineering

1. EXECUTIVE SUMMARY

This report documents the validation of the Shielding Attenuation Simulator V1.09. Version 1.09 adds a constraint-based material optimization layer above the previously validated V1.08 single-material comparison workflow.

The baseline attenuation model, NIST XCOM interpolation, point source inverse-square model, isotope source model, Geometric Progression (G-P) buildup equation, target-driven minimum-thickness calculations, and V1.08 material comparison calculations are unchanged. Therefore, this report focuses on the new engineering selection logic: mass-per-area calculation, relative cost indexing, hard constraint filtering, candidate status classification, objective based ranking, balanced normalization, and best design selection.

V1.09 allows the simulator to answer a new engineering question: among the materials that can satisfy a shared radiation target, which material is preferred after considering limits on thickness, mass per unit area, and relative cost? The optimizer distinguishes calculation failures from designs that calculate successfully but violate engineering requirements. Candidates are classified as ELIGIBLE, REJECTED, or FAILED, and only eligible candidates participate in ranking.

Validation confirmed that V1.09 correctly calculates engineering metrics, applies multiple simultaneous design constraints, preserves V1.08 failures and buildup warnings, selects the correct candidate for minimum thickness, minimum mass, minimum cost, and balanced objectives, and reports infeasible design problems without selecting an unacceptable material. The complete validation runner produced 215 PASS assertions and ended with “All validation tests passed.”

2. METHODOLOGY AND PHYSICS MODELS

2.1 Previously Validated Physics and Design Calculations

V1.09 does not introduce a new photon attenuation equation, source model, buildup equation, or minimum thickness solver. Each candidate begins as a V1.08 MaterialDesignCandidate generated by repeatedly applying the V1.07 single material minimum thickness pathway to the same source, detector distance, target, calculation search limit, and buildup setting.

The narrow-beam calculation remains based on Beer-Lambert attenuation using NIST XCOM mass attenuation coefficients converted to linear attenuation coefficients by density. Detector flux remains based on isotropic point-source geometric spreading. Isotope photon lines remain evaluated independently and summed at the source result level. When buildup is requested, each

candidate uses the existing single-layer G-P buildup aware design pathway or its documented narrow-beam fallback behavior.

2.2 Optimization Workflow

The top level `optimize_material_selection` function performs the following sequence:

- Run the V1.08 material comparison for all selected materials.
- Convert each passing V1.08 result into a V1.09 optimized candidate.
- Calculate mass per area and relative cost index per area.
- Apply all active hard engineering constraints.
- Classify each candidate as ELIGIBLE, REJECTED, or FAILED.
- Normalize metrics for eligible candidates when balanced optimization is used.
- Assign objective scores, rank eligible candidates, and select the best design.
- Preserve rejection reasons, calculation failure reasons, buildup fallbacks, and optimization warnings.

2.3 Mass per Unit Area

Because V1.09 does not yet define a complete shield geometry, it compares shielding mass using mass per unit area rather than total mass. For a homogeneous slab equivalent candidate:

$$m/A = \rho x$$

where:

- ρ is material density in g/cm³
- x is required thickness in cm
- m/A is mass per area in g/cm²

This metric is geometry independent and allows a physically meaningful comparison between dense thin materials and low-density thick materials.

2.4 Relative Cost Index per Area

V1.09 introduces an intentionally simplified relative cost model. Each material receives a positive, dimensionless relative cost multiplier I_C , interpreted on a comparative per unit mass basis. Ordinary concrete is assigned the baseline value 1.0. The cost metric is calculated as

$$C_A = (m/A) \cdot I_C$$

C_A is a relative cost index per area and is not a monetary unit. For the arithmetic validation case, lead uses $I_C = 3.0$. The relative cost library is intended to validate the optimization framework, not to represent current market prices. The program prints a warning whenever relative cost affects ranking or constraint filtering.

2.5 Hard Constraints and Candidate Status

V1.09 separates the numerical calculation search bound from engineering design constraints. The `calculation_max_thickness` parameter limits how far the lower level solver may search. A material that cannot meet the radiation target within that search range is FAILED. By contrast,

DesignConstraints are applied only after a valid required thickness has been calculated. A physically successful design that exceeds a maximum thickness, mass per area, or relative cost limit is REJECTED rather than FAILED.

The status definitions are:

- ELIGIBLE: the V1.08 calculation passed and every active engineering constraint is satisfied.
- REJECTED: the V1.08 calculation passed, but one or more engineering constraints are violated.
- FAILED: the underlying V1.08 shielding calculation did not produce a usable design within the calculation bounds or current model limitations.

All active constraints are evaluated independently. Therefore, a rejected material may preserve multiple rejection reasons instead of stopping at the first violation.

2.6 Optimization Objectives

V1.09 supports four objective modes. Lower objective values are preferred:

- Minimum thickness: rank by required shielding thickness.
- Minimum mass: rank by mass per unit area.
- Minimum cost: rank by relative cost index per area.
- Balanced: rank by a weighted combination of normalized thickness, mass, and relative cost.

Each single metric mode uses secondary deterministic tie-breakers, ending with the stable material key. This prevents output order from changing when candidates have equal primary values.

2.7 Balanced Normalization and Score

Thickness, mass per area, and relative cost cannot be added directly because they use different units and numerical scales. Balanced mode first applies min-max normalization among eligible candidates only:

$$N = \frac{(x - x_{min})}{(x_{max} - x_{min})}$$

where:

- x is the value for the material currently being evaluated.
- x_{min} is the best, or lowest, value among all eligible materials.
- x_{max} is the worst, or highest, value among all eligible materials.
- N is the normalized result, between 0 and 1.

Because the optimizer is minimizing thickness, mass, and cost:

- $N = 0$ means the material has the best eligible value.
- $N = 1$ means the material has the worst eligible value.
- A value between 0 and 1 shows where it falls between the best and worst.

The best eligible value becomes 0 and the worst becomes 1. If every eligible candidate has the same value for a metric, that normalized metric is assigned 0 because the metric does not distinguish the candidates. Rejected and failed designs are excluded so they cannot distort the ranking scale.

The weighted balanced score is:

$$S = \frac{w_t N_t + w_m N_m + w_c N_c}{w_t + w_m + w_c}$$

where:

- S is the final balanced optimization score.
- w_t is the thickness weight.
- w_m is the mass weight.
- w_c is the cost weight.
- N_t is the normalized required thickness.
- N_m is the normalized mass per unit area.
- N_c is the normalized relative cost index per area.

2.8 Command-Line Interface and Result Model

V1.09 adds a fourth command-line calculation mode for constraint based material optimization. The user selects a source, candidate materials, detector distance, target, calculation search limit, optional engineering constraints, optimization objective, optional balanced weights, and buildup setting.

The OptimizedDesignResult stores the complete candidate list, constraints, objective, optional weights, best candidate, selection reason, and warnings. The output table reports required thickness, mass per area, relative cost index per area, objective score, buildup status, and optimization status. Rejection reasons, calculation failures, buildup fallback warnings, and cost model warnings are printed below the table.

3. TEST CASE RESULTS

3.1 Engineering Metric Calculation Tests

The validation runner directly tests the new arithmetic helpers independently of the shielding solver. These tests confirm unit consistent mass per area calculation and correct application of the relative material cost index.

Test Case	Input Parameters	Expected Result	Actual Result	Status
1	Lead density = 11.34 g/cm ³ , thickness = 7.000 cm	m/A = 79.38 g/cm ²	79.38 g/cm ²	PASS

2	Lead $m/A = 79.38 \text{ g/cm}^2$, relative cost index = 3.0	$C_A = 238.14$	238.14	PASS
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3.2 Constraint Filtering and Rejection Tests

Constraint tests verify that V1.09 rejects successful designs for engineering reasons without converting them into calculation failures. They also verify that multiple simultaneous violations are preserved.

Test Case	Input Parameters	Expected Result	Actual Result	Status
3	Cs-137 candidate set; maximum design thickness = 10 cm	Water status = REJECTED	Water status = REJECTED	PASS
4	Rejected water candidate	At least one rejection reason stored	Rejection reason stored	PASS
5	Water only; maximum thickness = 10 cm; maximum mass/area = 50 g/cm^2	Status = REJECTED	Status = REJECTED	PASS
6	Same two constraint water case	More than one rejection reason stored	Two rejection reasons stored	PASS

3.3 Objective Ranking Tests

The optimizer was tested using a 1 Ci Cs-137 source, detector distance 100 cm, FluxTarget(100 photons/cm²/s), a six material comparison set, and buildup requested. With no hard constraints, each objective must select the minimum eligible value for its own metric.

Test Case	Input Parameters	Expected Result	Actual Result	Status
7	Objective = minimum_thickness	Smallest eligible thickness selected	Tungsten selected; 4.90817 cm	PASS
8	Objective = minimum_mass	Smallest eligible mass/area selected	Lead selected; 80.3803 g/cm^2	PASS
9	Objective = minimum_cost	Smallest eligible relative cost selected	Water selected; 22.8331	PASS

The different winners are consistent with the implemented objective definitions and relative-cost assumptions. Tungsten minimizes thickness, lead minimizes mass per area, and water minimizes the simplified relative cost index when no thickness constraint is active.

3.4 Balanced Optimization Tests

Balanced optimization was tested with weights 0.5 for thickness, 0.3 for mass, and 0.2 for relative cost. For every eligible candidate, the validation runner checks that the normalized score is not below 0 and not above 1. It then verifies that the selected candidate has the smallest score.

Test Case	Input Parameters	Expected Result	Actual Result	Status
10	Six eligible candidates; balanced weights = 0.5, 0.3, 0.2	Every score in [0, 1]	All six scores in [0, 1]	PASS
11	Balanced objective	Smallest score selected	Lead selected; score = 0.0308135	PASS
12	Lead normalized metrics	Weighted equation reproduces score	Calculated score = 0.0308135	PASS

3.5 Failure Preservation, Infeasibility, and Input Validation

The optimizer must preserve lower level failures, report infeasible constraint sets without selecting a material, and reject invalid optimizer inputs.

Test Case	Input Parameters	Expected Result	Actual Result	Status
13	Cs-137; lead and water; FluxTarget(1×10^{-30}); calculation limit = 0.1 cm	Failed V1.08 lead remains FAILED	Lead remains FAILED	PASS
14	All comparison materials; maximum design thickness = 0.1 cm	best_candidate = None	best_candidate = None	PASS
15	Same infeasible constraint case	Warning generated	No-eligible warning generated	PASS
16	DesignConstraints(max_thickness_cm = -1)	ValueError	ValueError	PASS
17	OptimizationWeights(0, 0, 0)	ValueError	ValueError	PASS

3.6 Manual Command-Line Constrained Optimization Example

A manual equivalent command-line case was executed using all nine available materials. The design problem used a 1 Ci Cs-137 source, 100 cm detector distance, target flux of 100 photons/cm²/s, calculation search limit of 100 cm, buildup requested, maximum design thickness of 20 cm, maximum mass per area of 120 g/cm², and the minimum mass objective.

Lead was selected because it had the lowest mass per area among the two eligible candidates. Tungsten was thinner, but its greater density produced a higher mass per area.

Material	Thickness (cm)	Mass/Area (g/cm ²)	Rel. Cost/Area	Buildup	Status
Lead	7.08821	80.3803	241.141	Yes	ELIGIBLE
Tungsten	4.90817	94.5020	2362.55	Yes	ELIGIBLE
Aluminum	58.3158	157.453	629.810	Yes	REJECTED
Concrete (Barite, Type BA)	28.7625	100.669	161.070	Fallback	REJECTED
Concrete (Ordinary)	62.0647	148.955	148.955	Yes	REJECTED
Copper	17.2589	154.640	1237.12	Yes	REJECTED
Polyethylene	94.6039	88.9277	177.855	Fallback	REJECTED
Tin	18.6007	135.971	951.799	Yes	REJECTED
Water	91.3324	91.3324	22.8331	Fallback	REJECTED

The constrained output demonstrates the distinction between efficiency metrics and feasibility. Water has a low mass per area and the lowest relative cost index, but it is rejected because it requires more than 20 cm. Copper and tin fit within 20 cm but exceed the 120 g/cm² mass limit. Ordinary concrete and aluminum violate both active constraints. Barite concrete and polyethylene violate the thickness limit. Lead and tungsten remain eligible, and lead wins the minimum mass objective.

3.7 All-Material Objective Comparison

The same unconstrained all nine material Cs-137 case was also run under all four optimization objectives. The selected material changes as expected when the design priority changes.

Objective	Selected Material	Winning Value	Interpretation
Minimum thickness	Tungsten	4.90817 cm	Smallest required physical thickness
Minimum	Lead	80.3803 g/cm ²	Smallest mass per unit area

mass			
Minimum relative cost	Water	22.8331	Lowest simplified relative cost index per area
Balanced (0.5/0.3/0.2)	Lead	0.0308135	Lowest weighted normalized score

3.8 Regression Preservation

The validation runner still includes all previously validated tests for attenuation, interpolation, source handling, buildup behavior, isotope calculations, target driven minimum thickness, fallback handling, and V1.08 material comparison. Running the uploaded V1.09 code produced 215 PASS assertions and no failures.

Test Case	Input Parameters	Expected Result	Actual Result	Status
18	V1.05 manual lead regression	Uncollided and buildup-corrected results unchanged	Regression tests PASS	PASS
19	V1.06 source and isotope tests	Line counts, photon rates, sums, and warnings valid	Source tests PASS	PASS
20	V1.07 minimum-thickness tests	Targets, bisection, buildup, and fallback valid	Design tests PASS	PASS
21	V1.08 material comparison tests	Candidate count, target, fallback, and failure behavior valid	Comparison tests PASS	PASS
22	Complete validation runner	All tests pass	215 PASS assertions; final success message	PASS

4. ASSUMPTIONS AND LIMITATIONS

1. Single-Material Optimization Only: V1.09 ranks homogeneous single material shields generated by the V1.08 comparison workflow. It does not optimize multilayer stacks.
2. Previously Validated Physics Reused: V1.09 validates engineering selection logic rather than introducing or independently revalidating new attenuation physics.

3. Mass per Area, Not Total Mass: The optimizer uses ρx in g/cm^2 because no complete shield geometry is defined. Total mass requires area or three-dimensional geometry.
4. Relative Cost Index, Not Currency: Cost values are simplified comparative assumptions. They are not dollars, market prices, supplier quotes, or procurement estimates.
5. Ordinary Concrete Baseline: The initial cost library assigns ordinary concrete an index of 1.0 and scales the other materials relative to that internal baseline.
6. Cost Dataset Is User-Editable: Changing a material cost index can change minimum cost and balanced rankings. The current report validates calculation behavior, not economic accuracy.
7. Hard Constraints Are Scalar Limits: V1.09 supports maximum thickness, maximum mass per area, and maximum relative cost index per area. It does not model structural, thermal, chemical, fabrication, toxicity, or regulatory constraints.
8. Calculation Limit and Design Constraint Are Different: A solver search failure produces FAILED, while a successful design that violates an engineering limit produces REJECTED.
9. Candidate-Set-Dependent Normalization: Balanced min-max scores depend on the eligible candidates included in a run. Adding, removing, or rejecting a material can change normalized scores.
10. Eligible Candidates Only Affect Balanced Scaling: Rejected and failed candidates are excluded from normalization so infeasible designs do not distort the competition among feasible designs.
11. Equal-Metric Normalization: When all eligible candidates have the same value for one metric, the normalized value for that metric is set to 0 for all candidates.
12. User-Selected Weights: Balanced results depend on user-specified thickness, mass, and cost weights. Weights represent preference and are not universal engineering constants.
13. Deterministic Tie-Breaking: Secondary metrics and material keys are used to produce repeatable rankings when primary objective values are equal.
14. No Uncertainty Propagation: Density, attenuation data, source intensity, calculated thickness, and cost indices are treated as deterministic inputs.
15. Single-Layer Homogeneous Buildup Only: G-P buildup remains supported only for supported single-layer homogeneous shields and within the 40-MFP validity range.
16. Mixed Modeling Quality Can Appear in One Table: Some candidates may use valid buildup aware thicknesses while unsupported or out of range candidates use documented narrow-beam fallback results.

17. Supported Buildup Materials: Lead, aluminum, ordinary concrete, water, tungsten, copper, and tin have G-P exposure buildup coefficient data in the simulator.
18. Unsupported Buildup Materials: Barite concrete and polyethylene remain available for narrow-beam calculations but do not have implemented G-P coefficient tables.
19. Exposure Buildup Coefficients: The implemented G-P tables are exposure buildup coefficients. The simulator does not yet distinguish photon flux, exposure response, air kerma, or dose response for final OpenMC comparison.
20. Point Source and One-Dimensional Geometry: The source is modeled as isotropic and shielding thickness lies along the source to detector path. Total thickness cannot exceed detector distance.
21. Selected Photon Lines: Isotope sources use selected major photon lines rather than complete decay spectra.
22. No Dose Conversion: V1.09 optimizes the currently implemented target quantities and does not report dose rate, exposure rate, air kerma, or effective dose.
23. No Monte Carlo Benchmark Yet: OpenMC verification and multilayer transport comparison are reserved for later versions.

5. CONCLUSION

The Shielding Attenuation Simulator V1.09 successfully adds constraint based material selection while preserving the previously validated attenuation, source, buildup, minimum thickness, and material comparison behavior.

Validation confirmed that the optimizer correctly calculates mass per area and relative cost index per area, separates calculation failures from engineering rejections, preserves multiple rejection reasons, and selects the minimum eligible candidate for thickness, mass, and relative cost objectives. Balanced optimization correctly normalizes eligible metrics, applies user defined weights, constrains scores to the expected range, and selects the smallest weighted score.

The constrained Cs-137 example demonstrated a useful engineering tradeoff. Tungsten produced the thinnest shield, but lead produced the lowest mass per area among designs satisfying both the 20 cm thickness limit and 120 g/cm² mass limit. Other materials were retained in the output with explicit reasons for rejection rather than disappearing from the comparison.

The complete V1.09 validation runner produced 215 PASS assertions, including all earlier regression tests. V1.09 therefore provides a stable decision layer for future tradeoff plots, scenario files, two layer optimization, OpenMC benchmarking, and the final shielded source design study.

References

- [1] Y. Harima, Y. Sakamoto, S. Tanaka, and M. Kawai, “Validity of the Geometric-Progression Formula in Approximating Gamma-Ray Buildup Factors,” *Nuclear Science and Engineering*, vol. 94, no. 1, pp. 24–35, Sep. 1986. doi:10.13182/nse86-a17113
- [2] D. K. Trubey, “New gamma-ray buildup factor data for point kernel calculations: ANS-6.4.3 standard reference data,” IAEA, <https://inis.iaea.org/records/0arzw-ez976/files/20014493.pdf> (accessed Jul. 10, 2026).
- [3] Angstrom Sciences, “Density of elements chart – Angstrom Sciences elements density table,” Angstrom Sciences, <https://angstromsciences.com/density-elements-chart> (accessed Mar. 26, 2026).
- [4] J. Colledge, M. Wilczek, and C. Raymond, “Inverse square law | radiology reference article,” Radiopaedia, <https://radiopaedia.org/articles/inverse-square-law?lang=us> (accessed Mar. 26, 2026).
- [5] “Shielding Radiation Alphas, Betas, Gammas and Neutrons,” U.S. NRC, <https://www.nrc.gov/docs/ML1122/ML11229A721.pdf> (accessed Mar. 26, 2026).
- [6] M. J. Berger et al., “XCOM: Photon Cross Sections Database,” NIST, <https://dx.doi.org/10.18434/T48G6X> (accessed Mar. 26, 2026).
- [7] National Nuclear Data Center, “NuDat 3 Glossary,” Brookhaven National Laboratory, <https://www.nndc.bnl.gov/nudat3/guide/glossary.html> (accessed Jul. 11, 2026).
- [8] International Atomic Energy Agency, “LiveChart of the Nuclides,” IAEA Nuclear Data Section, <https://nds.iaea.org/relnsd/vcharthtml/VChartHTML.html> (accessed Jul. 11, 2026).